

MCMC Audit — How the Fit Actually Ran

run 20260916_045356 · seed 20260914 · 2000 draws x 4 chains · target_accept 0.95 · max_treedepth 12

Every series below is also written to tables/mcmc_*.csv so the run can be audited without reopening the trace. The log-posterior trace is the closest analogue to a training-loss curve; ESS growth is the closest analogue to a learning curve.

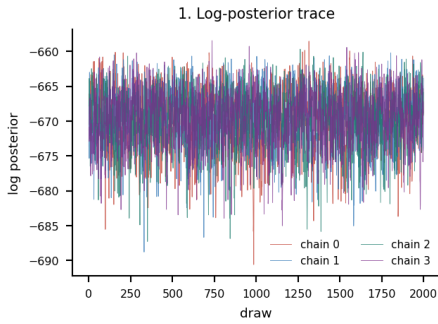
Convergence ledger

Parameter	mean	sd	R-hat	ESS bulk	ESS tail	MCSE/SD	Verdict
e_R0	1.935	0.032	1.0007	7784	5653	1.13%	PASS
mea_R0	8.947	0.877	1.0004	8064	4962	1.13%	PASS
flu_R0	3.028	2.82	1.0018	3478	1791	2.08%	PASS
n_R0	1.078	0.0214	1.0009	2466	3770	2.07%	PASS
n_attack_rate	0.1409	0.0337	1.0009	2466	3770	2.06%	PASS
log_intercept_e	-0.9979	0.0861	1.0003	7335	5848	1.17%	PASS
e_alpha_nb	2.16	0.625	1.0009	7258	5727	1.22%	PASS
mea_alpha_nb	0.9538	0.318	1.0006	7010	5776	1.24%	PASS
flu_alpha_nb	0.9213	0.32	1.0006	5612	4488	1.29%	PASS
class_re_sd	0.888	0.252	1.0002	2977	3893	1.81%	PASS
hyper_log_R0_mu	0.9376	0.376	1.0018	1608	2615	2.49%	PASS
hyper_log_R0_sd	0.7119	0.222	1.0000	3515	3768	1.65%	PASS

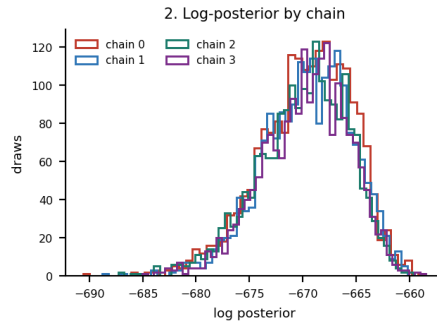
Sampler-state checks

Check	Value	Criterion	Verdict
Divergent transitions	0	0	PASS
Minimum E-BFMI	0.780	> 0.3	PASS
Draws at max tree depth	1.0%	< 5%	PASS
Mean acceptance	0.944	target 0.95	PASS
Rank-uniformity failures	0 of 12	0	PASS

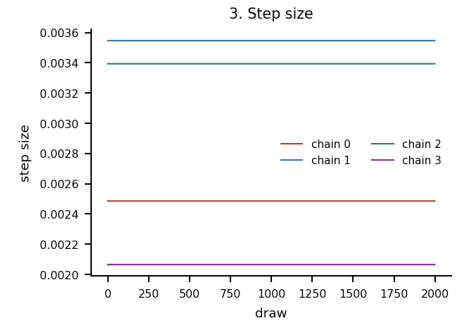
Sampler-process figures



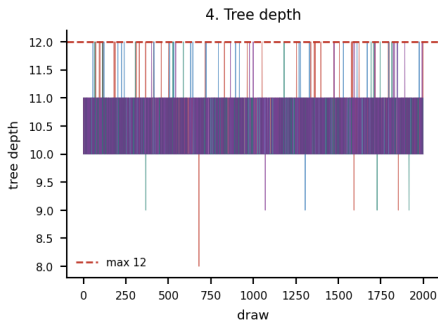
Log-posterior per draw — the MCMC equivalent of a training-loss curve. All chains overlap at the same level with no drift: none is stuck in a worse mode.



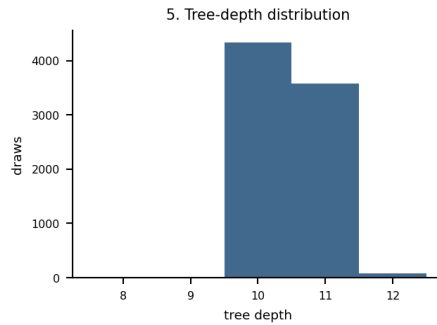
Log-posterior distribution per chain. Overlapping histograms confirm the chains are exploring the same region of parameter space.



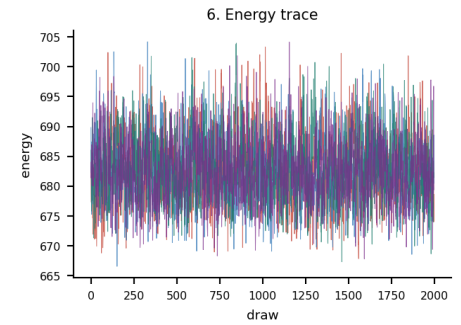
Tuned step size per chain. Flat across sampling means dual averaging settled during warm-up; chains landing at similar values means they found comparable geometry.



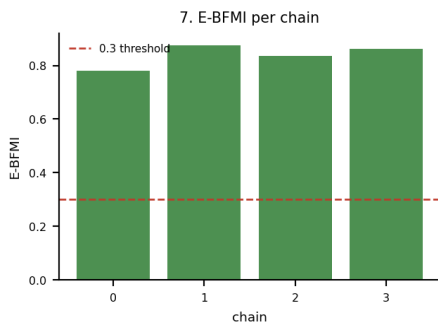
Leapfrog tree depth per draw. 1.0% of draws hit the ceiling of 12. Saturation is not an error but means the trajectory never turned around inside the budget, which costs efficiency.



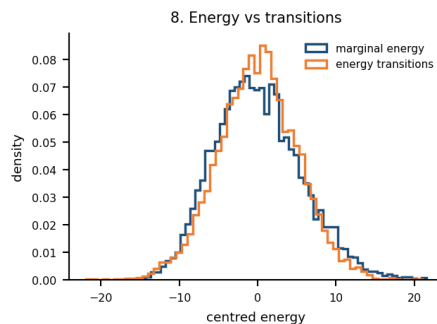
How often each tree depth was reached. Mass well below the ceiling means most trajectories terminated naturally.



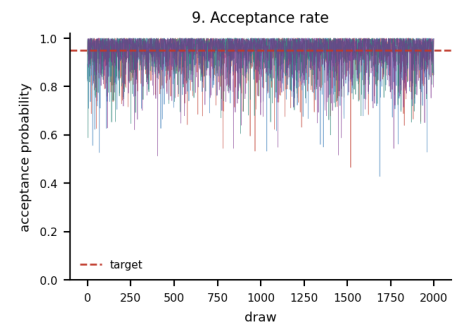
Hamiltonian energy per draw. Stable, overlapping traces mean the sampler is exploring the energy distribution rather than drifting.



Energy-based fraction of missing information per chain. Above 0.3 means the sampler is not being defeated by heavy tails (Betancourt 2018).

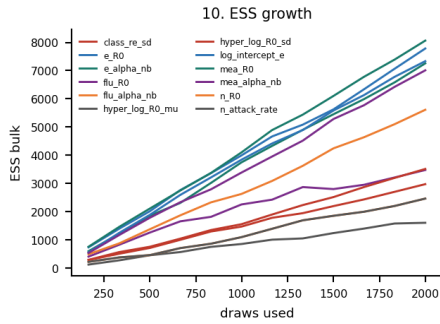


Marginal energy against the energy-transition distribution. Closely matching shapes is the visual form of a healthy BFM.

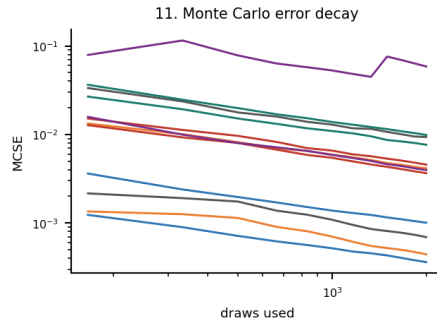


Acceptance probability per draw against the target of 0.95. Clustering at the target means the step size was tuned correctly.

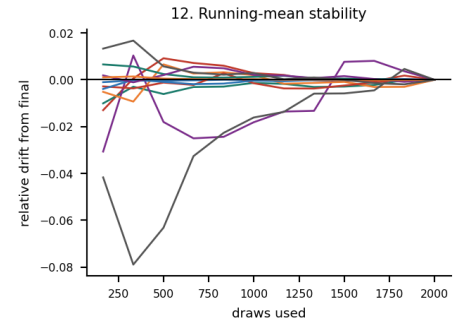
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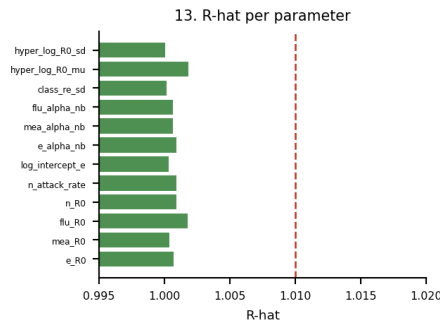
Effective sample size against chain length. A straight line through the origin means each extra draw buys the same information; flattening would mean longer runs no longer help.



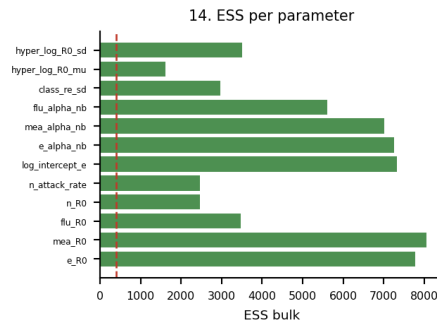
Monte Carlo error against chain length. Falling as one over the root of the draw count is the expected rate; departures indicate poor mixing.



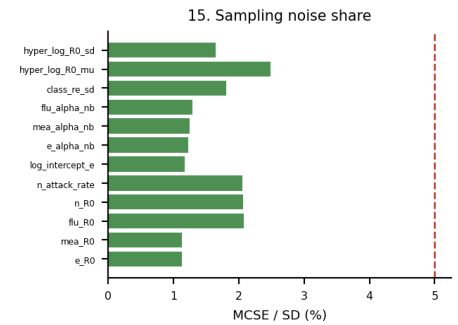
Running posterior mean as draws accumulate. Flat lines mean the estimate stopped moving long before the chain ended.



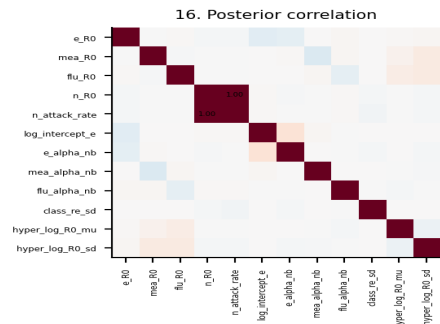
R-hat per parameter against the 1.01 target. Every bar below the line means all chains agree on the same distribution.



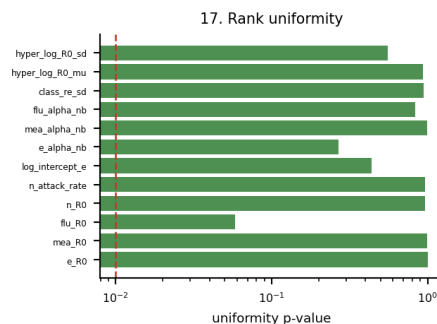
Effective sample size per parameter. Above 400 is the working threshold for a four-chain run.



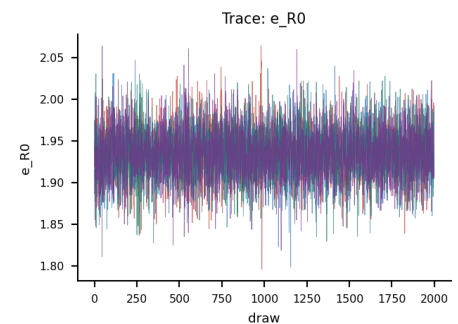
Monte Carlo error as a share of each parameter's own spread. Below 5% means sampling noise is negligible next to genuine uncertainty.



Posterior correlation between parameters. Strong off-diagonal values mark ridges where two parameters trade off and must move together.

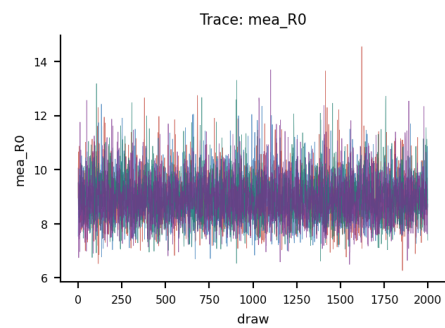


Rank-uniformity test per parameter (Vehtari et al. 2021). Passing means the chains agree about where the probability mass sits — a stricter check than R-hat alone.

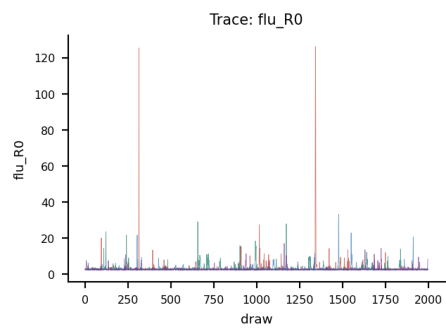


Chain trace for e_R0. Overlapping fuzzy caterpillars with no trend is what a converged parameter looks like.

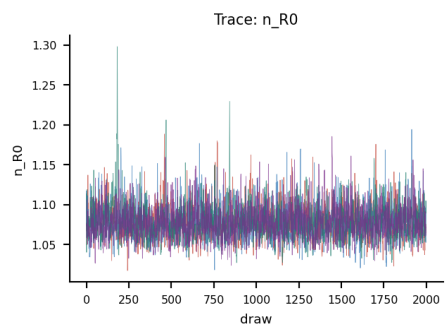
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Chain trace for mea_R0. Overlapping fuzzy caterpillars with no trend is what a converged parameter looks like.



Chain trace for flu_R0. Overlapping fuzzy caterpillars with no trend is what a converged parameter looks like.



Chain trace for n_R0. Overlapping fuzzy caterpillars with no trend is what a converged parameter looks like.